

4(a)(i)	0.3 ;	1
4(a)(ii)	<i>any four from:</i> 1 genetic drift ; 2 (drift as) frequency, fluctuates / decreases and increases / shows random changes / does not change markedly in just one direction ; 3 <u>directional</u> selection ; 4 from (generation) 2 to 7 / at first, T decreases as, T is selected against <u>and</u> from (generation) 7 to 13 / later, T increased as, T is selected for; 5 may be, stabilising selection / selection for heterozygotes ; 6 (stabilising selection) keeps relative allele frequencies roughly the same ;	4

Question	Answer	Marks
4(b)	any three from: 1 small(er) population ; 2 causes (genetic) bottleneck (effect) ; 3 chance events, have greater relative effect on / cause larger fluctuations in, allele frequency ; 4 more likely for one allele to, replace another / go extinct / become fixed ; 5 by chance T organisms, did not mate / pass on T allele or by chance offspring with T died ;	3
4(c)	<u>Hardy-Weinberg</u> ;	1

Question	Answer	Marks
----------	--------	-------